1) a)

	x	-2	
x	x^2	$-2x$	$+4$
$+1$	$+x$	-2	

$\frac{x^2-x+2}{x+1} = x - 2 + \frac{4}{x+1}$

$\therefore x = -1$ and $y = x - 2$

1) b)

	x	$+6$	
x	x^2	$+6x$	$+18$
-4	$-4x$	-24	

$\frac{x^2+2x-6}{x-4} = x + 6 + \frac{18}{x-4}$

$\therefore x = 4$ and $y = x + 6$

1) c)

	$3x$	-5	
x	$3x^2$	$-5x$	$+3$
$+2$	$+6x$	-10	

$\frac{3x^2+x-7}{x+2} = 3x - 5 + \frac{3}{x+2}$

$\therefore x = -2$ and $y = 3x - 5$

1) d)

	$-x$	$+3$	
x	$-x^2$	$+3x$	$+2$
$+3$	$-3x$	9	

$\frac{11-x^2}{x+3} = -x + 3 + \frac{2}{x+3}$

$\therefore x = -3$ and $y = -x + 3$

2) a)

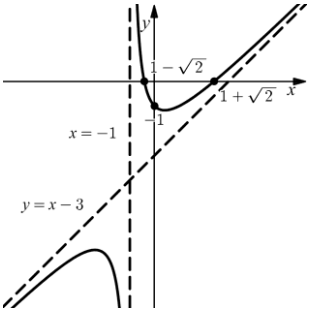
	x	-3	
x	x^2	$-3x$	$+4$
$+1$	$+x$	-3	

$\frac{x^2-2x-1}{x+1} = x - 3 + \frac{4}{x+1}$

Asymptotes: $x = -1$ and $y = x - 3$

When $x = 0$, $y = -1$

When $y = 0$, $x^2 - 2x - 1 = 0 \therefore x = 1 \pm \sqrt{2}$



2) b)

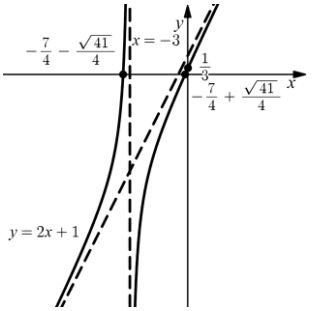
	$2x$	$+1$	
x	$2x^2$	$+x$	-2
$+3$	$+6x$	$+3$	

$\frac{2x^2+7x+1}{x+3} = 2x + 1 - \frac{2}{x+3}$

Asymptotes: $x = -3$ and $y = 2x + 1$

When $x = 0$, $y = \frac{1}{3}$

When $y = 0$, $2x^2 + 7x + 1 = 0 \therefore x = -\frac{7}{4} \pm \frac{\sqrt{41}}{4}$



2) c)

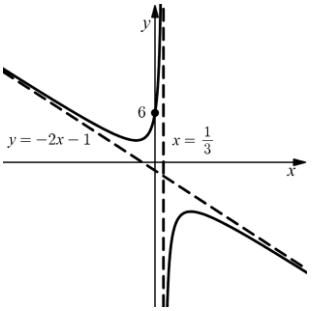
	$-2x$	-1	
$-3x$	$6x^2$	$+3x$	$+7$
$+1$	$-2x$	-1	

$\frac{6x^2+x+6}{1-3x} = -2x - 1 + \frac{7}{1-3x}$

Asymptotes: $x = \frac{1}{3}$ and $y = -2x - 1$

When $x = 0$, $y = 6$

When $y = 0$, $6x^2 + x + 6 = 0 \therefore$ since $b^2 - 4ac < 0$ then no real solutions



2) d)

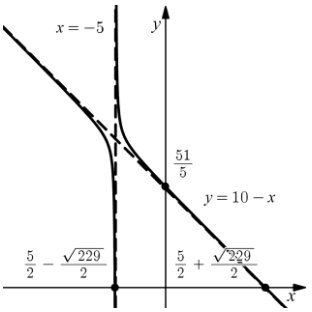
	$-x$	$+10$	
x	$-x^2$	$+10x$	$+1$
$+5$	$-5x$	$+50$	

$\frac{51+5x-x^2}{x+5} = 10 - x + \frac{1}{x+5}$

Asymptotes: $x = -5$ and $y = 10 - x$

When $x = 0$, $y = \frac{51}{5}$

When $y = 0$, $51 + 5x - x^2 = 0 \therefore x = \frac{5}{2} \pm \frac{\sqrt{229}}{2}$



3) a)

$y = 5 - 2x + \frac{k}{x-2}$ for some k

$0 = 5 - 0 + \frac{k}{0-2} \Rightarrow k = 10$

$y = 5 - 2x + \frac{10}{x-2} = \frac{5(x-2)-2x(x-2)+10}{x-2} = \frac{-2x^2+9x}{x-2}$

$\therefore a = -2$, $b = 9$, $c = 0$ and $d = -2$

3) b)

$y = x - 1 + \frac{k}{x+2}$ for some k

$\frac{1}{2} = 0 - 1 + \frac{k}{0+2} \Rightarrow k = 3$

$y = x - 1 + \frac{3}{x+2} = \frac{x(x+2)-1(x+2)+3}{x+2} = \frac{x^2+x+1}{x+2}$

$\therefore a = 1$, $b = 1$, $c = 1$ and $d = 2$

3) c)

$y = 2 - x + \frac{k}{x+1}$ for some k

$4 = 2 - 0 + \frac{k}{0+1} \Rightarrow k = 2$

$y = 2 - x + \frac{k}{x+1} = \frac{2(x+1)-x(x+1)+2}{x+1} = \frac{-x^2+x+4}{x+1}$

$\therefore a = -1$, $b = 1$, $c = 4$ and $d = 1$

3) d)

$y = 2x + 1 + \frac{k}{x-1}$ for some k

$-1 = 0 + 1 + \frac{k}{0-1} \Rightarrow k = 2$

$y = 2x + 1 + \frac{k}{x-1} = \frac{2x(x-1)+(x-1)+2}{x-1} = \frac{2x^2-x+1}{x-1}$

$\therefore a = 2$, $b = -1$, $c = 1$ and $d = -1$

4) a)

	ax	$-a$	
x	ax^2	$-ax$	$+a$
$+1$	$+ax$	$-a$	

$\frac{ax^2}{x+1} = ax - a + \frac{a}{x+1}$

Asymptotes: $x = -1$ and $y = ax - a$

4) b)

	ax	$+2a$	
x	ax^2	$+2ax$	$+4a$
-2	$-2ax$	$-4a$	

$\frac{ax^2}{x-2} = ax + 2a + \frac{4a}{x-2}$

Asymptotes: $x = 2$ and $y = ax + 2a$

4) c)

	ax	$a+b$	
x	ax^2	$(a+b)x$	$a+b$
-1	$-ax$	$-a-b$	

$\frac{ax^2+bx}{x-1} = ax + a + b + \frac{a+b}{x-1}$

Asymptotes: $x = 1$ and $y = ax + a + b$

4) d)

	ax	$-a^2$	

x	ax^2	$-a^2x$	$b+a^3$
$+a$	$+a^2x$	$-a^3$	

$\frac{ax^2+b}{x+a} = ax - a^2 + \frac{a^3+b}{x+a}$

Asymptotes: $x = -a$ and $y = ax - a^2$

4) e)

	$\frac{a}{b}x$	$\frac{ac}{b^2}$	
bx	ax^2	$+\frac{ac}{b}x$	$+\frac{ac^2}{b^2}$
$-c$	$-\frac{ac}{b}x$	$-\frac{ac^2}{b^2}$	

$\frac{ax^2}{bx-c} = \frac{a}{b}x + \frac{ac}{b^2} + \frac{\frac{ac^2}{b^2}}{bx-c}$

Asymptotes: $x = \frac{c}{b}$ and $y = \frac{a}{b}x + \frac{ac}{b^2}$

5) a)

$x = 1$

5) b)

	x	$k+1$	
x	x^2	$(k+1)x$	$k+1$
-1	$-x$	$-k-1$	

$\frac{x^2+kx+4}{x-1} = x + k + 1 + \frac{k+1}{x-1}$

Asymptote: $y = x + k + 1$

So $k + 1 = 3 \therefore k = 2$

6) a)

$x = 2$

6) b)

	$2x$	$4+k$	
x	$2x^2$	$(4+k)x$	$4+k$
-2	$-4x$	$-4-k$	

$\frac{2x^2+kx-3}{x-2} = 2x + k + 4 + \frac{4+k}{x-2}$

Asymptote: $y = 2x + k + 4$

So $k + 4 = 7 \therefore k = 3$

7) a)

	ax	$a + b$	
x	ax^2	$(a + b)x$	$4 + a + b$
-1	$-ax$	$-a - b$	

$\frac{ax^2+bx+4}{x-1} = ax + a + b + \frac{4+a+b}{x-1}$

Oblique asymptote: $y = 3x + 2$

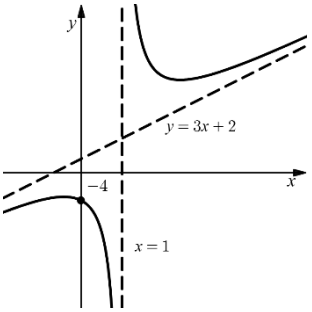
$\therefore a = 3$ and $b = -1$

7) b)

$x = 1$

7) c)

When $x = 0$, $y = -4$



8) a)

Find the values for which y exists:

$k = \frac{x^2+4x+7}{x+1} \Rightarrow k(x + 1) = x^2 + 4x + 7$

$x^2 + (4 - k)x + (7 - k) = 0$

For k to exist: $(4 - k)^2 - 4(1)(7 - k) \geq 0$

$16 - 8k + k^2 - 28 + 4k \geq 0$

$k^2 - 4k - 12 \geq 0$

$(k + 2)(k - 6) \geq 0$ as req.

8) b)

When $k = -2$:

$x^2 + (4+2)x + (7+2) = 0 \Rightarrow x^2 + 6x + 9 = 0 \Rightarrow (x+3)^2 = 0 \therefore x = -3$

When $x = -3$ then $(-3, 2)$

When $k = 6$:

$x^2 + (4-6)x + (7-6) = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x-1)^2 = 0 \therefore x = 1$

When $x = 1$ then $(1, 6)$

8) c)

	x	$+3$	
x	x^2	$+3x$	$+4$
$+1$	$+x$	$+3$	

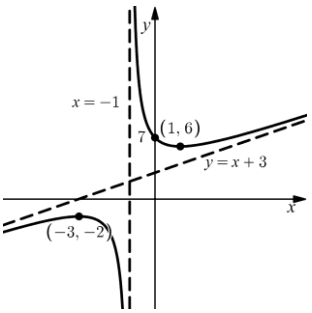
$\frac{x^2+4x+7}{x+1} = x + 3 + \frac{4}{x+1}$

Asymptotes: $x = -1$ and $y = x + 3$

8) d)

When $x = 0$, $y = 7$

When $y = 0$, $x^2 + 4x + 7 = 0$ has no real solutions.



9) a)

Find the values for which $y = k$ exists:

$k = \frac{-x^2+6x+12}{x+2} \Rightarrow k(x+2) = -x^2 + 6x + 12$

$x^2 + (k-6)x + (2k-12) = 0$

For k to exist: $(k-6)^2 - 4(1)(2k-12) \geq 0$

$k^2 - 12k + 36 - 8k + 48 \geq 0$

$k^2 - 20k + 84 \geq 0$

$(k-6)(k-14) \geq 0$ as req.

9) b)

When $k = 6$:

$x^2 + (6-6)x + (12-12) = 0 \Rightarrow x^2 = 0 \therefore x = 0$

When $x = 0$ then $(0, 6)$

When $k = 14$:

$x^2 + (14-6)x + (28-12) = 0 \Rightarrow x^2 + 8x + 16 = 0 \Rightarrow (x+4)^2 = 0 \therefore x = -4$

When $x = -4$ then $(-4, 14)$

9) c)

	$-x$	$+8$	
x	$-x^2$	$+8x$	-4
$+2$	$-2x$	$+16$	

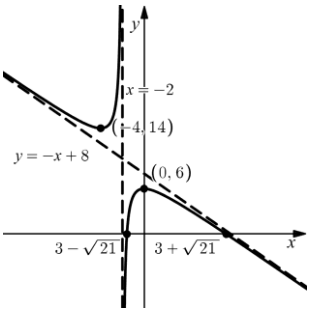
$\frac{-x^2+6x+12}{x+2} = -x + 8 - \frac{4}{x+2}$

Asymptotes: $x = -2$ and $y = -x + 8$

9) d)

When $x = 0$, $y = 6$

When $y = 0$, $-x^2 + 6x + 12 = 0 \therefore x =$



10) a)

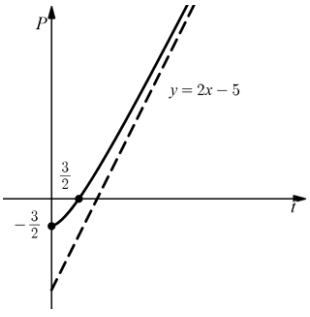
	$2t$	-5	
t	$2t^2$	$-5t$	$+7$
$+2$	$+4t$	-10	

$\frac{2t^2-t-3}{t+2} = 2t - 5 + \frac{7}{t+2}$

So $\frac{dP}{dt} = 2t - 5 + \frac{7}{t+2}$

When $t = 0$, $\frac{dP}{dt} = -\frac{3}{2}$

$2t^2 - t - 3 = 0 \therefore t = -1$ or $t = \frac{3}{2}$



10) b)

Initially $\frac{dP}{dt} < 0$ and the population is decreasing, then when $t = \frac{3}{2}$ then the growth rate becomes zero before it increases, tending towards an accelerating growth of $2t$

10) c)

$P = t^2 - 5t + 7\ln(t+2) + c$

10) d)

When $t = 0$, $P = 4$

$4 = 0 - 0 + 7\ln 2 + c \therefore c = 4 - 7\ln 2$

$P = t^2 - 5t + 7\ln(t+2) + 4 - 7\ln 2$

10) e)

When $t = 4$, $P = 7.69$ thousand residents

11) a)

	ax	$b - 3a$	
x	ax^2	$(b-3a)x$	$7 + 9a - 3b$
$+3$	$+3ax$	$3b - 9a$	

$\frac{ax^2+bx+7}{x+3} = ax + b - 3a + \frac{7+9a-3b}{x+3}$

Oblique asymptote: $y = x + 1$

$\therefore a = 1$ and $b = 4$

11) b)

$x = -3$

11) c)

Find the values for which $y = k$ exists:

$k = \frac{x^2+4x+7}{x+3} \Rightarrow k(x+3) = x^2 + 4x + 7$

$x^2 + (4-k)x + (7-3k) = 0$

For k to exist: $(4-k)^2 - 4(1)(7-3k) \geq 0$

$16 - 8k + k^2 - 28 + 12k \geq 0$

$k^2 + 4k - 12 \geq 0$

$(k+6)(k-2) \geq 0$

When $k = 2$:

$x^2 + 2x + 1 = 0 \Rightarrow (x+1)^2 = 0 \therefore x = -1$

When $x = -1$ then $(-1, 2)$

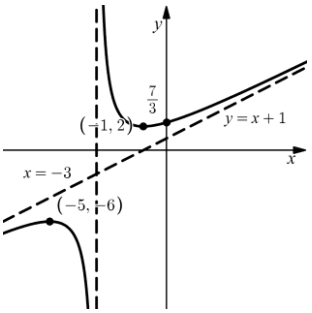
When $k = -6$:

$x^2 + 10x + 25 = 0 \Rightarrow (x+5)^2 = 0 \therefore x = -5$

When $x = -5$ then $(-5, -6)$

11) d)

When $x = 0$, $y = \frac{7}{3}$



12) a)

	x	-2	
x	x^2	$-2x$	$+4$
$+2$	$+2x$	-4	

$\frac{x^2}{x+2} = x - 2 + \frac{4}{x+2}$

Asymptotes are $x = -2$ and $y = x - 2$

12) b)

Find the values for which $y = k$ exists:

$k = \frac{x^2}{x+2} \Rightarrow k(x+2) = x^2$

$x^2 - kx - 2k = 0$

For k to exist: $(-k)^2 - 4(1)(-2k) \geq 0$

$k^2 + 8k \geq 0$

$k(k+8) \geq 0$

When $k = 0$:

$x^2 = 0 \ \therefore x = 0$

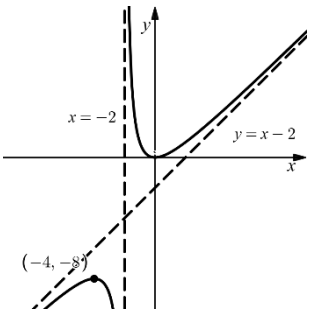
When $x = 0$ then $(0, 0)$

When $k = -8$:

$x^2 + 8x + 16 = 0 \Rightarrow (x+4)^2 = 0 \therefore x = -4$

When $x = -4$ then $(-4, -8)$

12) c)



12) d)

$$\pi\int_0^2\left(\frac{x^2}{x+2}\right)^2\,\mathrm{d}x=\pi\int_0^2\frac{x^4}{(x+2)^2}\,\mathrm{d}x$$

Let $u=x+2$ then $\frac{du}{\mathrm{d}x}=1$

When $x=0$, $u=2$, when $x=2$, $u=4$ and $x^4=(u-2)^4$:

$$\pi\int_0^2\frac{x^4}{(x+2)^2}\,\mathrm{d}x=\pi\int_2^4\frac{(u-2)^4}{u^2}\,\mathrm{d}u=\pi\int_2^4\frac{u^4-8u^3+24u^2-32u+16}{u^2}\,\mathrm{d}u$$

$$=\pi\int_2^4u^2-8u+24-\frac{32}{u}+16u^{-2}\,\mathrm{d}u=\pi\left[\frac{1}{3}u^3-4u^2+24u-32\ln u-\frac{16}{u}\right]_2^4$$

$$=\pi\left[\left(\frac{64}{3}-64+96-32\ln 4-4\right)-\left(\frac{8}{3}-16+48-32\ln 2-8\right)\right]$$

$$=\pi\left[\frac{68}{3}+32(\ln 2-\ln 4)\right]$$

$$=\pi\left[\frac{68}{3}+32\ln\frac{1}{2}\right]$$

$$=\pi\left[\frac{68}{3}-32\ln 2\right]\text{ as req.}$$

13) a)

	$3x$	$+1$	
x	$3x^2$	$+x$	$+1$
$+1$	$+3x$	$+1$	

$\frac{3x^2+4x+2}{x+1}=3x+1+\frac{1}{x+1}$

Asymptotes are $x=-1$ and $y=3x+1$

13) b)

Find the values for which $y=k$ exists:

$k=\frac{3x^2+4x+2}{x+1}\Rightarrow k(x+1)=3x^2+4x+2$

$3x^2+(4+k)x+(2+k)=0$

For k to exist: $(4+k)^2 - 4(3)(2+k) \geq 0$

$k^2 - 4k - 8 \geq 0$

When $k = 2 + 2\sqrt{3}$:

$3x^2 + \left(6 + 2\sqrt{3}\right)x + \left(4 + 2\sqrt{3}\right) = 0 \therefore x = -1 - \frac{\sqrt{3}}{3}$

When $x = -1 - \frac{\sqrt{3}}{3}$ then $\left(-1 - \frac{\sqrt{3}}{3}, -2 - 2\sqrt{3}\right)$

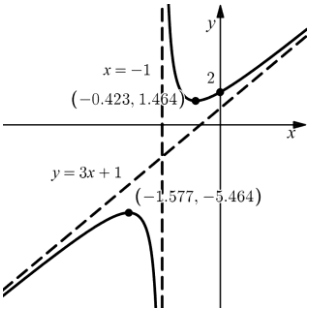
When $k = 2 - 2\sqrt{3}$:

$3x^2 + \left(6 - 2\sqrt{3}\right)x + \left(4 - 2\sqrt{3}\right) = 0 \ \ \therefore x = -1 + \frac{\sqrt{3}}{3}$

When $x = -1 + \frac{\sqrt{3}}{3}$ then $\left(-1 + \frac{\sqrt{3}}{3}, -2 + 2\sqrt{3}\right)$

$\therefore (-1.577,\ -5.464)$ and $(-0.423,\ 1.464)$

13) c)



13) d)

$\frac{1}{4-1}\int_1^4 \left(3x + 1 + \frac{1}{x+1}\right)\,\mathrm{d}x$

$= \frac{1}{3}\left[\frac{3}{2}x^2 + x + \ln(x+1)\right]_1^4$

$= \frac{1}{3}\left[(24 + 4 + \ln 5) - \left(\frac{3}{2} + 1 + \ln 2\right)\right]$

$= \frac{1}{3}\left[\frac{51}{2} + (\ln 5 - \ln 2)\right]$

$= \frac{51}{6} + \frac{1}{3}\ln\frac{5}{2}$

13) e)

$\frac{51}{6} + \frac{1}{3}\ln\frac{5}{2} \approx 8.81$ (hundreds of pounds)

The mean value represents the average production cost per batch when output varies between 1 and 4 batches per day. Over this interval, the company can expect the cost per batch to be approximately £881 on average.

14) a)

	x^2	$-2x$	$+4$	
x	x^3	$-2x^2$	$+4x$	-12
$+2$	$+2x^2$	$-4x$	$+8$	

$\frac{x^3-4}{x+2} = x^2 - 2x + 4 - \frac{12}{x+2}$

Asymptotes are $x = -2$ and $y = x^2 - 2x + 4$

14) b)

	x^2	$+5x$	$+10$	
x	x^3	$+5x^2$	$+10x$	$+21$
-2	$-2x^2$	$-10x$	-20	

$\frac{x^3+3x^2+1}{x-2} = x^2 + 5x + 10 + \frac{21}{x-2}$

Asymptotes are $x = 2$ and $y = x^2 + 5x + 10$

14) c)

	$-x^2$	$+x$	-1	
x	$-x^3$	$+x^2$	$-x$	$+2$
$+1$	$-x^2$	$+x$	-1	

$\frac{1-x^3}{x-1} = -x^2 + x - 1 + \frac{2}{x+1}$

Asymptotes are $x = -1$ and $y = -x^2 + x - 1$

15) a)

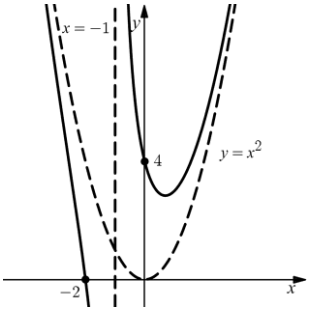
	x^2	$+0x$	$+0$	
x	x^3	$+0x^2$	$+0x$	$+4$
$+1$	$+x^2$	$+0x$	$+0$	

$\frac{x^3+x^2+4}{x+1} = x^2 + \frac{4}{x+1}$

Asymptotes: $x = -1$ and $y = x^2$

When $x = 0$, $y = 4$

When $y = 0$, $x^3 + x^2 + 4 = 0 \Rightarrow (x+2)(x^2 - x + 2) = 0 \therefore x = -2$



15) b)

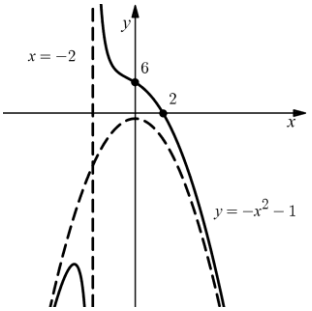
	$-x^2$	$+0x$	$+1$	
x	$-x^3$	$+0x^2$	$+x$	$+15$
$+3$	$-3x^2$	$+0x$	$+3$	

$\frac{18+x-3x^2-x^3}{x+3} = -x^2 + 1 + \frac{15}{x+3}$

Asymptotes: $x = -3$ and $y = -x^2 + 1$

When $x = 0$, $y = 6$

When $y = 0$, $18 + x - 3x^2 - x^3 = 0 \Rightarrow -(x-2)(x^2+5x+9) = 0 \therefore x = 2$



15) c)

	x^2	$-4x$	$+0$	
x	x^3	$-4x^2$	$+0x$	$+12$
$+1$	$+x^2$	$-4x$	$+0$	

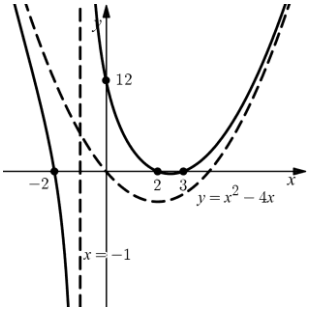
$\frac{x^3-3x^2-4x+12}{x+1} = x^2 - 4x + \frac{12}{x+1}$

Asymptotes: $x = -1$ and $y = x^2 - 4x$

When $x = 0$, $y = 12$

When $y = 0$, $x^3 - 3x^2 - 4x + 12 = 0 \Rightarrow (x-3)(x-2)(x+2) = 0$

$\therefore x = -2, x = 2, x = 3$



15) d)

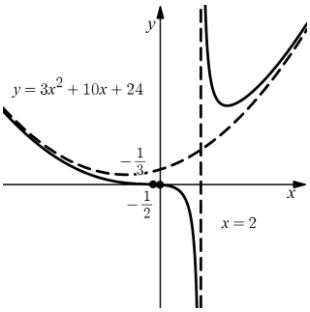
	$3x^2$	$+10x$	$+24$	
x	$3x^3$	$+10x^2$	$+24x$	$+49$
-2	$-6x^2$	$-20x$	-48	

$\frac{3x^3+4x^2+4x+1}{x-2} = 3x^2 + 10x + 24 + \frac{49}{x-2}$

Asymptotes: $x = 2$ and $y = 3x^2 + 10x + 24$

When $x = 0$, $y = -\frac{1}{2}$

When $y = 0$, $3x^3 + 4x^2 + 4x + 1 = 0 \Rightarrow (3x + 1)(x^2 + x + 1) = 0 \therefore x = -\frac{1}{3}$



16) a)

$y = x^2 + 3x + 11 + \frac{k}{x-3}$ for some k

$\frac{7}{3} = 11 + \frac{k}{0-3} \Rightarrow k = 26$

$y = x^2 + 3x + 11 + \frac{26}{x-3} = \frac{x^2(x-3)+3x(x-3)+11(x-3)+26}{x-3} = \frac{x^3+2x-7}{x-3}$

$\therefore a = 1, b = 0, c = 2, d = -7$ and $p = -2$

16) b)

$y = -x^2 - x + 1 + \frac{k}{x+1}$ for some k

$3 = 1 + \frac{k}{0+1} \Rightarrow k = 2$

$y = -x^2 - x + 1 + \frac{2}{x+1} = \frac{-x^2(x+1)-x(x+1)+1(x+1)+2}{x+1} = \frac{-x^3-2x^2+3}{x+1}$

$\therefore a = -1, b = -2, c = 0, d = 3$ and $p = 1$

16) c)

$y = x^2 - x + 5 + \frac{k}{x+4}$ for some k

$\frac{21}{4} = 5 + \frac{k}{0+4} \Rightarrow k = 1$

$y = x^2 - x + 5 + \frac{1}{x+4} = \frac{x^2(x+4)-x(x+4)+5(x+4)+1}{x+4} = \frac{x^3+3x^2+x+21}{x+4}$

$\therefore a = 1, b = 3, c = 1, d = 21$ and $p = 4$

16) d)

$y = -x^2 + x + \frac{k}{x-2}$ for some k

$-1 = \frac{k}{0-2} \Rightarrow k = 2$

$y = -x^2 + x + \frac{2}{x-2} = \frac{-x^2(x-2)+x(x-2)+2}{x-2} = \frac{-x^3+3x^2-2x+2}{x-2}$

$\therefore a = -1, b = 3, c = -2, d = 2$ and $p = -2$

17) a)

	ax^2	$-ax$	$+a$	
x	ax^3	$-ax^2$	$+ax$	$-a$
$+1$	$+ax^2$	$-ax$	$+a$	

$\frac{ax^3}{x+1} = ax^2 - ax + a - \frac{a}{x+1}$

Asymptotes: $x = -1$ and $y = ax^2 - ax + a$

17) b)

	ax^2	$+2ax$	$+4a$	
x	ax^3	$+2ax^2$	$+4ax$	$+8a$
-2	$-2ax^2$	$-4ax$	$-8a$	

$\frac{ax^3}{x-2} = ax^2 + 2ax + 4a + \frac{8a}{x-2}$

Asymptotes: $x = 2$ and $y = ax^2 + 2ax + 4a$

17) c)

	ax^2	$(a+b)x$	$a+b$	
x	ax^3	$(a+b)x^2$	$(a+b)x$	$a+b$
-1	$-ax^2$	$-(a+b)x$	$-a-b$	

$\frac{ax^3+bx^2}{x-1} = ax^2 + (a+b)x + a + b + \frac{a+b}{x-1}$

Asymptotes: $x = 1$ and $y = ax^2 + (a+b)x + a + b$

17) d)

	ax^2	$-a^2x$	a^3+b	
x	ax^3	$-a^2x^2$	$(a^3+b)x$	$-a^4-ab$
$+a$	$+a^2x^2$	$-a^3x$	a^4+ab	

$\frac{ax^3+bx}{x+a} = ax^2 - a^2x + a^3 + b - \frac{a^4+ab}{x+a}$

Asymptotes: $x = -a$ and $y = ax^2 - a^2x + a^3 + b$

17) e)

	$\frac{a}{b}x^2$	$+\frac{ac}{b^2}x$	$+\frac{ac^2}{b^3}$	
bx	ax^3	$+\frac{ac}{b}x^2$	$+\frac{ac^2}{b^2}x$	$+\frac{ac^3}{b^3}$
$-c$	$-\frac{ac}{b}x^2$	$-\frac{ac^2}{b^2}x$	$-\frac{ac^3}{b^3}$	

$\frac{ax^3}{bx-c} = \frac{a}{b}x^2 + \frac{ac}{b^2}x + \frac{ac^2}{b^3} + \frac{\frac{ac^3}{b^3}}{bx-c}$

Asymptotes: $x = \frac{c}{b}$ and $y = \frac{a}{b}x^2 + \frac{ac}{b^2}x + \frac{ac^2}{b^3}$

18) a)

$x = 1$

18) b)

	x^2	$(1+k)x$	$1+k$	
x	x^3	$(1+k)x^2$	$+(1+k)x$	$-1+k$
-1	$-x^2$	$-(1+k)x$	$-1-k$	

$\frac{x^3+kx^2-2}{x-1} = x^2 + (1+k)x + 1 + k + \frac{-1+k}{x-1}$

Asymptote: $y = x^2 + (1+k)x + 1 + k$

So $k + 1 = 5 \therefore k = 4$

19) a)

$x = -3$

19) b)

	x^2	$(-k-3)x$	$+3k+9$	

x	x^3	$(-k-3)x^2$	$(3k+9)x$	$-9k+20$
$+3$	$+3x^2$	$(-3k-9)x$	$+9k+27$	

$\frac{x^3-kx^2+47}{x+3} = x^2+(-k-3)x+3k+9+\frac{-9k+20}{x+3}$

Asymptote: $y = x^2+(-k-3)x+3k+9$

So $-k-3=-5 \therefore k=2$